

Optimising participant grouping methods in bilingualism studies

Insights from eye-tracking data

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This research addresses two major challenges in studying second language acquisition and bilingualism: reducing overlap in predictor variables and correctly classifying participants into language proficiency levels. Too many relevant predictors can harm statistical analysis due to an increased chance of overlap, known as multicollinearity. To tackle this, we use Principal Component Analysis (PCA) on selected predictors to identify proficiency indicators, combining the length of stay in the UK and language test scores. Additionally, traditional methods, especially IELTS-based proficiency classifications, often miss subtle differences in language skills, particularly when they fail to consider how long participants have been exposed to the target language. We counter this by using non-hierarchical Cluster Analysis (NCA) for a grounded, data-driven way of detecting distinct language proficiency groups. This new approach is demonstrated on a dataset of eye movements from reading tasks, collected from Chinese–English bilinguals in the UK.

Keywords: principal component analysis, non-hierarchical cluster analysis, generalised additive mixed models, eye movements in reading, L2 proficiency classification

1. Introduction

In quantitative research, particularly in bilingualism studies, participant grouping methods and the issue of multicollinearity often present significant challenges. Accurately classifying participants into proficiency levels is essential for understanding cognitive processes, yet traditional methods can oversimplify language abilities (Hulstijn, 2012). When key variables overlap — such as vocabulary test scores and length of language exposure — the resulting multicollinearity can

distort statistical models, leading to unreliable inferences about language proficiency's true effects (Jia et al., 2002; Wurm & Fisicaro, 2014).

To overcome these challenges, advanced statistical techniques like Principal Component Analysis (PCA) and non-hierarchical Cluster Analysis (NCA) offer data-driven solutions. PCA reduces dimensionality by transforming correlated variables into independent components, isolating the distinct contributions of proficiency and exposure variables (Divjak & Milin, 2020). NCA further refines participant grouping by identifying natural divisions in proficiency levels, allowing for a more nuanced understanding of bilingualism. Unlike traditional methods such as IELTS band cut-offs, which often fail to capture subtle differences in proficiency and impose broad group boundaries, these approaches integrate multiple indicators to provide a more precise and sensitive categorisation. This study applies these methods to a dataset on second language (L2) reading, addressing the limitations of traditional grouping strategies. We illustrate our approach to grouping participants using data that captures bilingual reading strategies through eye-tracking, a technique that allows us to track and analyse eye movements to gain insights into cognitive processes during reading. By sorting participants based on their second language proficiency, particularly in reading, we can better understand their patterns of eye movement in bilingual reading. In the following sections, we will first review relevant literature on bilingualism and eye-tracking research, setting the stage for our study. This review will be followed by a detailed explanation of how our new method for classifying participants, combined with eye-tracking data, can provide new insights into bilingual reading and bilingualism more generally.

Eye-tracking technology, which involves monitoring eye movements to understand how individuals interact with text, has emerged as a powerful tool in psycholinguistic research. By capturing data on where and how long a reader fixates on specific words or phrases, researchers can infer the cognitive processes underlying reading behaviour (Rayner, 1998). This method is particularly valuable in bilingualism studies, where it helps uncover differences in how bilinguals process language compared to monolinguals (Dussias, 2010). Eye-tracking research uses key measures to study reading behaviour and cognitive processing. First fixation duration (FFD) measures the time a reader's eyes first fixate on a word, reflecting the initial effort needed for word recognition (Rayner, 1998; Schotter et al., 2012). Total reading time (TRT) captures the total time spent on a word or text segment, indicating overall cognitive load. In bilingual research, longer TRT values suggest greater difficulty, often linked to language proficiency and exposure (Whitford & Titone, 2012). Eye movement studies have been pivotal in understanding bilingual reading processes. For example, Cop et al. (2015) compared monolingual and bilingual reading of a novel, revealing distinct patterns

that underscore the cognitive strategies employed by bilinguals during language processing. Additionally, Whitford and Titone (2012) have shown that second-language experience modulates first- and second-language word frequency effects, providing insights into how bilinguals process text differently depending on their proficiency in each language. Using eye-tracking, Kroll and Bialystok (2013) showed that bilinguals adjust their reading strategies based on language proficiency and context. Expanding on this, the present study employs empirically informed proficiency groups to predict bilingual reading patterns. Specifically, it investigates how to effectively group participants in bilingualism research using self-reports, vocabulary tests, and L2 exposure measures to capture the nuanced impact of proficiency on reading behaviour.

1.1 Critical predictors in bilingual reading and the problem of multicollinearity

The exploration of bilingual reading has uncovered a multitude of factors influencing language processing in individuals who can function in two languages. Among these, cognitive strategies, proficiency, linguistic exposure and immersion have been identified as significant contributors to how bilinguals navigate reading tasks in their second language (Chaouch-Orozco et al., 2021; Hofweber et al., 2016; Kroll & Bialystok, 2013). In bilingual reading, cognitive strategies refer to deliberate mental processes, such as inferencing, selective attention, and metacognitive monitoring, used to enhance comprehension and problem-solving (Bialystok, 2011; Mishra, 2018). These strategies help bilingual readers manage dual-language activation and reconcile linguistic differences, thereby improving both linguistic processing and cognitive control (Kroll et al., 2012). Proficiency, in bilingualism research, refers to an individual's ability to use a language effectively across different contexts, which influences their processing efficiency and reading comprehension (Hulstijn, 2012). Linguistic exposure encompasses the amount and quality of contact with a language, shaping bilinguals' vocabulary and syntactic knowledge (Linck et al., 2009). Immersion denotes sustained interaction in a second-language environment, fostering automaticity and contextual understanding essential for reading fluency (Dussias & Sagarra, 2007).

Given the intricate nature of bilingual language processing, where numerous factors interact in complex ways, traditional linear models may fall short in capturing these subtleties. Previous studies have highlighted the non-linear relationships between word-level variables and cognitive processing during reading (Kliegl et al., 2006). However, these studies often relied on polynomial functions in linear mixed models, which may not fully encapsulate the complexity of bilingualism. To address this, our study utilises Generalised Additive Mixed Models

(GAMMs), which provide a more robust framework for examining the non-linearities inherent in bilingual language processing. Recent work has highlighted GAMMs as a flexible and powerful approach for handling non-linear data in bilingualism research (Coretta & Casillas, 2024). GAMMs allow for a dynamic and comprehensive analysis, accounting for the diverse factors such as the age of L2 acquisition, L2 exposure, and cultural context, all of which play a crucial role in shaping language proficiency and processing.

Our study introduces an innovative methodology that examines a broad spectrum of proficiency indicators, moving beyond traditional single-dimension classifications like IELTS band cut-offs. By synthesising data through PCA and NCA, we develop a data-driven categorisation of language proficiency. This approach integrates multiple indicators, including vocabulary scores, reading proficiency, and immersion duration, addressing the oversimplification and multicollinearity challenges inherent in traditional methods.

IELTS-based cut-offs create broad group boundaries, often missing subtle distinctions in L2 proficiency. Traditional methods overlook the complex interplay of exposure, proficiency, and cognitive processing. Our PCA- and NCA-based approach refines classification using multiple indicators, offering a more flexible, data-driven alternative to rigid test-based groupings. Second, by uncovering non-linear relationships and group-specific processing strategies, our approach provides a more detailed understanding of how proficiency influences L2 reading. A comparison of results from both methods is included in the supplementary materials. While this paper focuses on English reading data, participants are bilingual, as the full study (He, 2024) also incorporates their Chinese reading data, reinforcing their classification as bilingual readers.

Recent literature has started to unravel the complex array of factors influencing bilingual reading, offering insights into both distinct and combined effects. Research by Schotter et al. (2014) has demonstrated that individuals with larger vocabularies tend to have fewer fixations on common words while focusing more on unfamiliar ones, indicating active vocabulary learning. This aligns with the cognitive strategies bilinguals use during language processing, highlighting the importance of vocabulary proficiency as a critical factor in bilingual reading. Similar studies such as Hulstijn (2012), Kroll and Bialystok (2013), and Marian and Fausey (2006) have deepened our understanding of bilingualism, underscoring the necessity for advanced methodological approaches. These works underline the significance of various factors, including language skills, proficiency levels, and the contexts in which languages are learned and utilised. However, as the body of research grows, so does the complexity of understanding the interrelationships among these factors, and the challenge in teasing them apart, particularly given the issue of multicollinearity — a statistical phenomenon where

predictor variables in a multiple regression model are highly correlated, affecting the reliability of estimates and, thus, distorting the inferences about the true effect of each predictor on the dependent variable (Jia et al., 2002; Ramanujan & Weekes, 2020; Schroeder et al., 2017; Wurm & Fisicaro, 2014). The interrelated nature of variables such as proficiency, immersion duration, and cognitive strategies presents substantial challenges, complicating the isolation of each factor's unique contributions to the bilingual reading process.

A promising approach to overcoming the challenge of multicollinearity involves the application of Principal Component Analysis (PCA), as exemplified by a study of Divjak and Milin (2020). Their study dealt with the interrelations among variables in language processing research, where factors like pattern learning ability and linguistic skills often intertwine. By employing PCA, they transformed correlated variables into a smaller number of uncorrelated principal components, each representing distinct aspects of the data's variance. This method allowed them to reduce data dimensionality, ensuring the new predictors were independent. Consequently, Divjak and Milin (2020) could isolate the unique contributions of different dimensions of statistical learning ability to language processing, providing deeper insights into the cognitive underpinnings of language acquisition and usage.

Acknowledging the above considerations, this paper aims to explore the complexities of bilingual reading by examining critical predictors and their interactions, specifically addressing multicollinearity. Our investigation stresses the importance of refined participant grouping strategies, responding to the methodological gaps identified in previous literature. The need for empirical studies employing sophisticated grouping methods to accurately capture the dynamics of bilingual experiences becomes evident.

1.2 L2 reading models and developmental trajectories

Second language (L2) reading is shaped by multiple factors, including lexical retrieval mechanisms, language exposure, and strategic processing. As proficiency develops, key reading behaviours – such as saccades, fixations, and regressions – tend to reflect greater automaticity in lexical access and more efficient processing routines (Rayner, 1998). Early-stage L2 readers often exhibit longer fixation durations and more frequent regressions, as they rely on slower, more effortful lexical retrieval (Whitford & Titone, 2012). However, as proficiency increases, readers demonstrate shorter fixations, fewer regressions, and smoother saccadic movements, indicating more automated word recognition (Schotter et al., 2012).

Language exposure plays a significant role in shaping these developmental trajectories. Greater exposure to the target language, especially in immersive environments, accelerates the automatization of word recognition and reduces the cognitive load during reading (Kroll & Bialystok, 2013). Similarly, the use of reading strategies – such as contextual guessing and inferencing – varies with proficiency, with advanced readers employing more efficient strategies that streamline processing and comprehension (Hulstijn, 2012).

The predictors in this study – such as proficiency level, exposure, and vocabulary size – are expected to influence these reading behaviours. As lexical retrieval becomes more automatic with increased exposure and proficiency, we anticipate smoother reading patterns, shorter fixations on familiar words on familiar words, and more rapid transitions between words. The incorporation of these predictors into our analysis allows for a more nuanced understanding of how bilinguals' reading behaviour evolves as they progress towards greater proficiency.

1.3 Participant grouping methods in bilingualism research

In bilingualism research, participant grouping often relies on standardised tests like IELTS, TOEFL, DELF, or HSK. While useful as benchmarks, these tests often oversimplify language proficiency by failing to capture its full range (Aizawa et al., 2020). Self-reported questionnaires provide additional insights but are prone to biases such as over- or under-estimation and lack of objectivity (Tempelaar et al., 2020; Tomoschuk et al., 2019). Hybrid models, combining standardised tests with self-reports, offer a more balanced approach. For instance, Whitford and Joannis (2021) combined the LEAP-Q with objective measures like Word Reading and Pseudoword Decoding tests for a comprehensive assessment. Dynamic assessment frameworks further enhance evaluations by tailoring them to individual learning paths through interactive feedback (Poehner, 2008; Lantolf & Poehner, 2011). Although resource-intensive, these methods provide nuanced insights into linguistic competence and language learning trajectories (Poehner, 2008; Shabani, 2023).

Eye-movement measures like FFD and TRT are valuable for studying cognitive processes in bilingual reading (Cop et al., 2015; Schotter et al., 2012; Whitford & Titone, 2012). This study focuses on FFD to explore how proficiency influences bilingual reading behaviours, as it reflects the effort required for word recognition during reading. Eye-tracking enables a detailed investigation of bilingual reading strategies and lexical access processes. By employing eye-tracking in this study, we aim to contribute to this line of research, using FFD to capture the nuances

of bilingual reading behaviours. FFD was selected based on its ability to provide detailed insights into word recognition.

1.4 This study

Building on the discussion of critical predictors in bilingual reading, such as cognitive strategies, proficiency, and linguistic exposure, and the evaluation of participant grouping methods, we now turn to the research question of this study: How can we effectively group participants in bilingualism studies using a mixture of self-reported data, vocabulary tests, and L2 exposure measures to accurately reflect the varying levels of bilingual language proficiency? This question explores the effectiveness of the new categorisation method in bilingual reading studies, especially regarding eye movement analysis. This study aims to develop and validate a new method for categorising participants. The proposed approach will use different types of data, including self-reports, vocabulary tests, and measures of L2 exposure, to form a detailed classification system that better represents the different levels of bilingual proficiency.

To illustrate our approach, we analyse data from an eye-tracking study and show how analysing eye movement patterns in bilingual reading, using our new grouping method, provides different and more detailed insights into bilingual language processing compared to traditional grouping methods. Note that the full description of the eye tracking reading study can be found in He (2024). He (2024) investigates how Chinese–English bilinguals process English reading through eye-tracking, focusing on cognitive mechanisms such as lexical access, word recognition, and reading fluency. The study involved L1 Chinese reader learners of English, residing in the UK, and L1 English readers, with the key measure being FFD captured using the EyeLink Portable Duo system. The research explores the impact of factors like word frequency, word length, and L2 proficiency on eye movements. The findings indicate that L2 readers, especially those with lower proficiency, exhibit longer fixation times and distinct processing strategies compared to L1 readers. The dissertation contributes to the understanding of bilingual reading behaviours and provides insights into how language proficiency and exposure influence reading efficiency.

In the next section, we describe our methodology, including participant demographics, data collection tools, and experimental protocols, with detailed explanations of participant selection processes, language history questionnaires, vocabulary assessments, and the specifics of the reading materials and eye-tracking technology utilised.

2. Methods

2.1 Participants

40 L1 Chinese readers who were L2 learners of English (14 male and 26 female, with a mean age of 23.3, $SD=2.7$) and 40 L1 English readers (17 male and 23 female, with a mean age of 21.5, $SD=5.2$) over the age of 18 were recruited as participants. All participants had normal or corrected-to-normal vision.

2.2 Language history questionnaires and vocabulary test

We utilised an adapted version of the Language History Questionnaire (Li et al., 2006) to assess the bilingual or second language acquisition backgrounds of our participants, aligning with methodologies validated by Park and Ziegler (2014). The questionnaire collected comprehensive demographic and language proficiency information from Chinese participants; it was simplified for English L1 speakers by excluding questions related to IELTS scores, lengths of stay in the UK, and experiences of learning English. The scores obtained from the IELTS, and vocabulary tests serve as independent variables that indicate participants' language proficiency. These variables will be used to assess their influence on eye-movement behaviour during the reading task. Basic L2 proficiency levels in English, including means and standard deviations for the key variables, were as follows: speaking ($M=6$, $SD=0.62$), listening ($M=7$, $SD=0.58$), reading ($M=7$, $SD=0.73$), writing ($M=6.0$, $SD=0.51$), and overall proficiency ($M=6.5$, $SD=0.46$). Detailed questionnaire content is available in Appendix 1A and 1B.

To gauge participants' vocabulary proficiency, we employed a computerised adaptation of the vocabulary subtests from the Nelson-Denny Reading Test (Brown et al., 1993). The Nelson-Denny Reading Test is a well-established standardised assessment of adult reading skills, as noted by previous studies such as Hersch and Andrews (2012). This task included 30 items consisting of a target word and five alternative words or short phrases, from which participants selected the option that most closely corresponded to the target's meaning. For example, one item presented the target word *toxin* with five possible answers: (a) poison; (b) weed; (c) failure; (d) belief; (e) disease.

2.3 Reading stimuli

The English text materials consisted of 29 short excerpts (5 to be used as practice trials) sourced from online news articles. The selection of these specific texts was driven by the objective of ensuring the readability of the materials. To achieve

this, we utilised the Flesch Reading Ease Score, aiming for texts that would be accessible yet engaging for our audience. The average score of our selected reading stimuli was 50, which corresponds to a readability level suitable for 10th to 12th graders. This level was deliberately chosen to accommodate the English proficiency of L2 readers, specifically Chinese learners of English, to avoid making the materials overly difficult. Examples of reading stimuli utilised in the experiments can be found in Appendix 2 for reference. A comprehensive list of reading stimuli can be found in the supplementary materials.

The excerpts were 44 to 98 words long ($M=70.7$, $SD=12.9$) and were shown in a single paragraph, fitting one display presentation. We randomly added a yes/no comprehension question after some paragraphs to make sure participants read the materials carefully. In total, every participant answered 6 comprehension questions. An example of a text and its accompanying comprehension question can be found in Appendix 3.

2.4 Apparatus

For the eye-tracking phase of the study, participants' eye movements were recorded with an EyeLink Portable Duo system (Version: 6.10.01). We used head-stabilised monocular recording, for the dominant eye, in pupil-only mode. The eye tracking system sampled at a rate of 500 Hz. Participants were seated 45 cm away from the (24-inch) display screen in a comfortable height-adjustable chair that was positioned at a fixed distance from the screen. The reading materials were displayed with black letters on a white background, employing Courier New font in Normal style, and a font size of 18. To enhance readability and minimise visual strain, we employed double vertical spacing between lines of text. The presentation of single excerpts and English questions was arranged in the vertical position, utilising a screen resolution of 1024*768 pixels.

2.5 Procedure

Following identification of the dominant eye and completing calibration and validation procedures, participants were prompted to initiate the practice trial. The calibration process involved a nine-point calibration, a standard method for reading multi-line text. This method facilitated accurate mapping of participants' eye movements and enabled precise eye tracking during the experiment. Participants were given prior notice that comprehension questions would be part of the study. The comprehension questions were answered with 93% average accuracy by English participants and 95% by Chinese participants. After the paragraph reading experiment, both groups completed the vocabulary test, followed by the

questionnaire. Full versions of the questionnaires for the Chinese and English participants can be found in Appendix 1A and Appendix 1B.

2.6 Eye-Movement Variables

In this study, the dependent variable (DV) is the First Fixation Duration (FFD), which captures the time spent on the first fixation when encountering a word, indicating the initial cognitive effort required for word recognition.

The independent variables (IVs) include participant-level variables, such as language proficiency (measured through IELTS scores and vocabulary test scores) and language exposure (duration of stay in the UK), as well as word-level variables, including word frequency and word length. We defined individual words as Interest Areas (IAs) for analysis, and FFDs were measured for the current word (word N), the preceding word (word N-1), and the following word (word N+1). These IVs will be used to assess how factors like language proficiency, exposure, word length, and word frequency affect the DV, FFD. Further details on data preparation are provided in Section 2.7.

2.7 Data preparation

Using the software Data Viewer (4.3.210) we exported the data, removing words that were not fixated as we focus on fixation durations on words that received at least one fixation. We also removed FFDs shorter than 60 ms or longer than 1000 ms from analysis because such extreme values are likely to be associated with artifacts, blinks, or data recording errors rather than meaningful cognitive processes related to reading (Ashby et al., 2005; Yan et al., 2010; Yen et al., 2009).

The complete English reading study dataset contains three main groups of variables: besides the eye-movement measure FirstFixDur, we included the vocabulary test scores and the results of the questionnaires. Additionally, we also included the following random factors, controls, and numeric predictors (covariates).

The independent variables used in the analysis include WordLength and Log-WordFreq for the current, previous, and following words, as well as participant-specific variables such as proficiency (based on IELTS and vocabulary scores) and exposure (time spent in the UK). These predictors are used to model their impact on FFD.

Random factors

We introduced Participant (CP₁ to CP₄₀, and EP₁ to EP₄₀), Story, and Word as random effect factors. Additionally, we combined the latter two variables into a

single random factor to reflect the fact that a word was part of a story (text). The model with these two random factors combined was by far superior to the next best model with the two random effects as separate terms (Decrease in Residual Deviance = 638.39, $p < .001$, Decrease in AIC = 232.6).

Controls

We introduced a variable named *Word_Story_Order*¹, to control for potential sequential effects, which refer to the order in which events occur and their timing, i.e., temporal effects. This variable combines the sequence number of texts (*TrialIndex*, ranging from 1 to 24) and the position of words within a given text (*IA*) into a single identifier, enabling us to examine the potential effects of word order within the narrative context. For example, a word positioned third in the eleventh text is represented as 11003, combining its text and word position. This approach enables us to examine order effects that may influence durational and other response measures. For example, changes in reading speed near a text's conclusion (e.g., end-of-story effect; Just & Carpenter, 1980) can be observed in a smoothed curve plotting order (X-axis) against duration (Y-axis).

Covariates

The main numerical predictors in our study were the length (number of letters) and frequency of the fixated word (*N*), the previous word (*N*−1), and the next word (*N*+1). We used these to examine whether and how the nearby words – i.e., words near the focal point that can greatly influence reading efficiency (Heister et al., 2012; Kliegl et al., 2006) – affect eye movements of L2 readers. For the word frequency counts we relied on the British National Corpus (BNC Consortium, 2007).

Data merging was performed using the *dplyr* package (Wickham et al., 2023) in R (version 4.3.2; R Core Team, 2023). The merged dataset from the English reading study included 139,121 eye-movement observations from 79 participants². Next, we conducted descriptive and preliminary analyses, removing extreme values and outliers to refine the dataset. Additionally, we applied variable transformations as needed to improve data suitability for statistical modelling, as suggested by Baayen and Milin (2010). The Box-Cox transformation was used to normalise FFD, a method recently highlighted as effective in bilingualism (Keating, 2024). After this cleaning process, 80,191 observations remained for further analysis.

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1. The *Word_Story_Order* variable combines the sequence number of texts and word positions within a text to control for potential temporal effects on fixation durations.
 2. Note that we removed the data of the one Chinese participant due to major data missing.

3. Data analysis and results

3.1 Clustering Chinese participants

Recall that Chinese participants reported their latest IELTS scores (including IELTS overall score, IELTS reading score, IELTS listening score, IELTS writing score and IELTS speaking scores), and durations of their stay in the UK (MonthUK). Besides, participants did an English vocabulary size test. This gave us data that is correlated, as summarised Figure 1. Figure 1 displays a correlation matrix of four key variables: the number of months spent in the UK (MonthUK), vocabulary size (VocabScore), reading proficiency measured by the IELTS (ReadingIELTS), and overall IELTS score (TotalIELTS). The correlation coefficients in the matrix range from -1 to $+1$, where positive values indicate a direct relationship between the variables, and negative values signify an inverse relationship. A value of zero implies no linear correlation between the variables.

The diagonal of the matrix includes histograms and density plots representing the univariate distributions of each variable. These plots offer a visual overview of how the scores for each variable are distributed across participants.

The matrix reveals several key relationships. A significant positive correlation exists between VocabScore and ReadingIELTS ($r = 0.41, p < .05$), and a very strong positive correlation is observed between ReadingIELTS and TotalIELTS ($r = 0.89, p < .001$), indicating a strong link between reading proficiency and overall IELTS performance. However, MonthUK shows only weak, non-significant correlations with the other variables, suggesting that the length of time spent in the UK does not directly predict improvements in vocabulary or reading proficiency.

In summary, the correlations depicted in Figure 1 suggest that while the duration of stay in an English-speaking environment (MonthUK) has minimal direct impact on language proficiency, vocabulary size and reading ability are closely related to overall English proficiency, as measured by the IELTS.

Correlation analysis

We ran Spearman's rank correlation test, as it is distribution-free (i.e., no normality/symmetry assumption), to examine the relationships amongst the variables. Results presented in Figure 1 indicate that the correlation between the IELTS overall score (TotalIELTS) and the IELTS reading score (ReadingIELTS) is highly significant ($\rho = 0.89, S = 1114.98, p < .001$). There are also other significant correlations as indicated in Figure 1.

While these correlations demonstrate important relationships between variables, they also raise concerns regarding multicollinearity, where predictor variables are highly correlated. Multicollinearity can skew the results of statistical

analyses by inflating standard errors, potentially leading to inaccurate conclusions about the significance of these predictors. It is therefore essential to address these correlations in subsequent analyses to ensure that the statistical models accurately capture the relationships between the variables without introducing bias or unreliability.

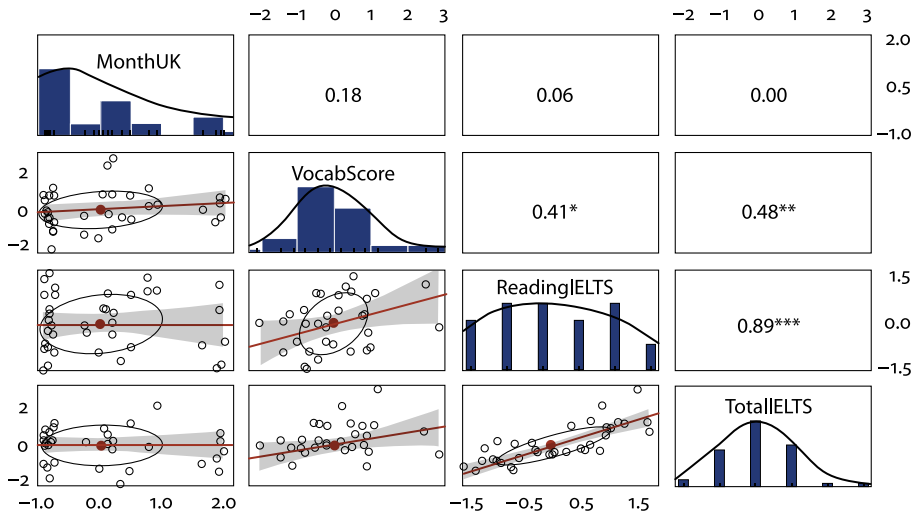


Figure 1. Correlations between four variables indicating language proficiency and L2 exposure

Principal component analysis

To tackle the issue of multicollinearity and to ensure a focused analysis on variables pivotal for this research, we implemented Principal Component Analysis (PCA) on essential variables: ReadingIELTS and VocabScore, and MonthUK. Principal Component Analysis (PCA) is a dimensionality reduction technique that transforms correlated variables into a set of uncorrelated components. This method is particularly valuable in addressing multicollinearity, where predictor variables are highly correlated, as it reduces the risk of inflated standard errors in regression models. The primary advantage of PCA is that it summarises complex data into principal components that represent the variance in the data. By doing so, it reduces the dimensionality of the data while retaining the most important features, which allows for a more focused and reliable analysis.

To minimise multicollinearity and maintain relevance to our study on eye movement measures, we excluded certain variables, such as TotalIELTS. This PCA process revealed common and independent components representative of the participants' L2 skills.

For the final PCA, we retained ReadingIELTS, VocabScore, and MonthUK as our variables of interest. We utilised Varimax rotation to achieve a “simple structure” of mutually orthogonal (uncorrelated) components, following the guidance of the Kaiser-Guttman criterion (Kaiser, 1960) and confirmation through the Scree test (Cattell, 1966). Varimax rotation was applied to the PCA to simplify the loadings and make the components more interpretable. This rotation method maximises the variance of squared loadings of factors across variables, producing what is called a ‘simple structure,’ where each variable loads highly on one principal component and minimally on others. All PCA and subsequent analyses were performed using the R statistical software (R Core Team, 2023). The stats package was used for both the PCA and non-hierarchical clustering (k-means) to ensure accuracy and reproducibility in the analyses. This resulted in two varimax-rotated components that together explained 81.3% of the total variance.

Table 1. Principal component analysis of three variables

Variable	RC1	RC2	<i>h</i> ²	<i>u</i> ²	<i>com</i>
MonthUK		0.97	0.94	0.06	1.0
VocabScore	0.80		0.71	0.28	1.2
ReadingIELTS	0.85		0.75	0.25	1.1

Note. *h*² = Communality; *u*² = Uniqueness (1 – *h*²); *com* = Complexity (number of components contributing to the variable). Loadings below an absolute value of 0.3 are not presented.

Table 1 summarises the results of the principal component analysis (PCA). The *h*² values (communality) represent the proportion of variance in each variable explained by the extracted components. The *u*² values (uniqueness) represent the proportion of variance not explained by the components (calculated as 1 – *h*²). The *com* values (complexity) reflect the number of components required to account for the variance of a given variable. Table 1 reveals that the first principal component significantly correlates with VocabScore and ReadingIELTS, indicating these variables heavily influence this component. The second component is strongly linked with MonthUK, highlighting its unique contribution to capturing the duration of stay in the UK. These components elucidate underlying data patterns, with the first reflecting English linguistic abilities and the second, the experience length in the UK. Figure 2 illustrates the loadings plot of the first two principal components from the PCA which visually presents these relationships and components. Dim1 and Dim2 correspond to the first and second dimensions of the PCA space, capturing the largest and second-largest sources of variance in the dataset. The *cos*² values indicate the quality of representation for each variable

on these dimensions, with higher values reflecting better alignment with the component structure.

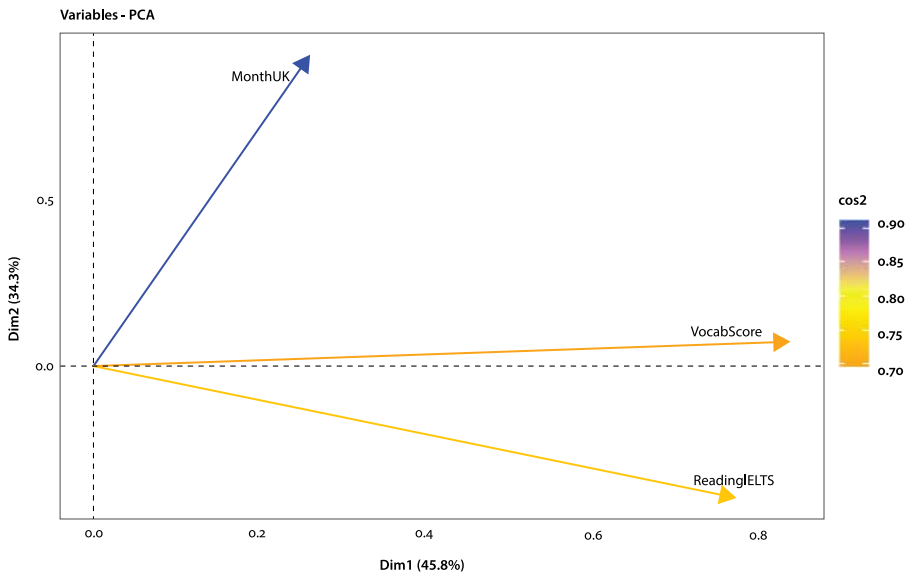


Figure 2. The plot of loadings for the first two principal components of a PCA. *Note.* Dim1 represents the primary dimension of variance, while Dim2 captures additional variance. The *cos2* values indicate the quality of representation for each variable within the two-dimensional factor space.

Next, we used scores on two principal components as our new variables of L2 proficiency and exposure of Chinese learners of English: PC.VocabAndReading and PC.MonthUK.

Subsequently, we applied non-hierarchical Cluster Analysis (NCA) to classify participants based on their principal component scores in R (R Core Team, 2023) using the *kmeans* function from the *stats* package, which is part of the base R distribution. NCA is a clustering method that organises data by grouping participants based on similarities in their scores, without requiring a predefined hierarchy of clusters. This method works by partitioning data into a specified number of clusters, minimising the variance within each cluster. For our analysis, we used the *k-means* clustering method, which aims to create homogenous groups by iteratively assigning participants to clusters that minimise the differences within each cluster. This approach is well-suited for uncovering natural groupings within complex datasets like ours. Figure 3 visualises two clusters of participants projected onto the surface of the two previously extracted Principal Components. The clustering analysis divided participants into two distinct groups, termed

‘Upper Intermediate’ and ‘Advanced’, based on their scores in PCA. The two principal components capture two dimensions: (1) Vocabulary and reading proficiency (RC1) and (2) Length of stay in the UK (RC2). The ‘Advanced’ group tends to score higher on both principal components, indicating greater proficiency and longer immersion in English-speaking environments. In contrast, the ‘Upper Intermediate’ group demonstrates lower proficiency and shorter stays in the UK

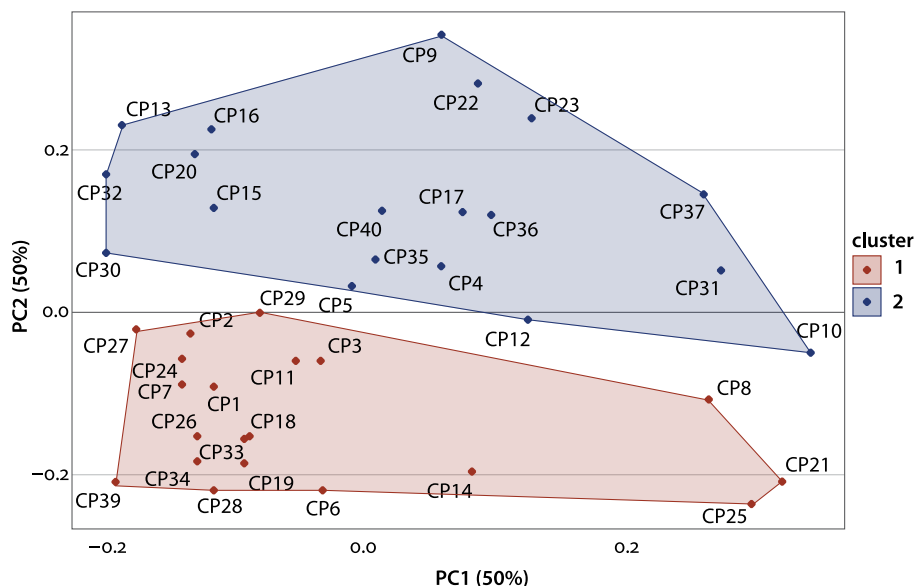


Figure 3. Clustering analysis result based on 39 Chinese participants

Interestingly, as Figure 3 indicates, PC.MonthUK (Y-axis) seems to be doing most of the work in separating natural groups of Chinese learners of English, with contribution from PC.VocabAndReading, which captures the downward tilt of both groups, indicating that the shorter time of immersion (PC.MonthsUK) can be (mildly) compensated for with higher values of PC.VocabAndReading. Note that utilising an IELTS cut-off of 6.5 would have resulted in a skewed distribution, categorising only 12 participants as C1 and disproportionately assigning 27 to B2 (He et al., 2022). Preliminary analyses using the IELTS-based categorisation revealed no significant differences between the two groups, nor between these groups and L1 readers, in terms of lexical-level effects and durations. This outcome underscores the necessity for a more nuanced approach in evaluating language competence and the role of immersion, to authentically represent the complexities of language acquisition in bilingual individuals. Such an imbalanced

grouping risks compromising the study's robustness, potentially skewing results and reducing the reliability of conclusions drawn.

Based on the result of the PCA and the NCA, we divided Chinese participants into 2 groups with 19 in the Advanced group and 20 in the Upper Intermediate group; see table in Appendix 4 for detailed grouping information of Chinese participants. The Cluster Analysis revealed two distinct groups. The Advanced group ($N=19$) had significantly higher scores on the vocabulary test ($M=52.1$) and IELTS reading score ($M=7.8$) compared to the Upper Intermediate group ($N=20$), whose mean vocabulary score was 45.3 and mean IELTS reading score was 6.5. Additionally, participants in the Advanced group had a longer average stay in the UK (25.5 months) compared to the Upper Intermediate group (18.2 months). These distinctions demonstrate that the cluster analysis captured meaningful differences in both language proficiency and language exposure.

In contrast, using the traditional IELTS-based grouping would have resulted in an uneven distribution of participants, with 12 participants classified as 'Advanced' and 27 as 'Upper Intermediate.' Notably, this division did not align with meaningful differences in vocabulary size or length of stay in the UK, suggesting that the cluster-based grouping offers a more accurate representation of participants' language proficiency and exposure. Moreover, the cluster analysis identified key differences in cognitive processing, as observed through eye-tracking measures, which were not detectable when using the IELTS-based classification.

The detailed comparison of group characteristics, as well as the more balanced distribution of participants into meaningful categories, underscores the advantage of using data-driven methods such as cluster analysis. This approach better reflects the complexity of bilingual proficiency, as evidenced by the distinct reading patterns observed between groups in our subsequent analysis of eye-tracking data.

Table 2. Descriptive statistics for participant groups

Grouping method	Mean vocabulary score	Mean reading score	Mean months in the UK
B2 (IELTS Cut-off)	10.2 ($SD=2.1$)	6.5 ($SD=0.7$)	12.5 ($SD=10.6$)
C1 (IELTS Cut-off)	13.6 ($SD=1.9$)	7.8 ($SD=0.6$)	23.4 ($SD=11.5$)
Upper Intermediate (PCA-based)	9.9 ($SD=2.3$)	6.6 ($SD=0.7$)	11.1 ($SD=8.6$)
Advanced (PCA-based)	13.4 ($SD=2.2$)	7.7 ($SD=0.8$)	22.9 ($SD=12.4$)

As shown in Table 2, the PCA-based clustering revealed distinct groupings with statistically significant differences in vocabulary scores and reading proficiency compared to the IELTS-based groupings. For example, the Advanced group (PCA-based) had significantly higher vocabulary scores ($M=13.4$, $SD=2.2$) and reading scores ($M=7.7$, $SD=0.8$) compared to the Upper Intermediate group ($M=9.9$, $SD=2.3$ and $M=6.6$, $SD=0.7$, respectively). The IELTS cut-off approach (B2 vs. C1) showed less nuanced separation of proficiency levels. Additionally, the PCA-based method also captured the effect of language exposure (i.e., months spent in the UK) more precisely.

We conducted independent-samples t -tests to evaluate whether the groups formed by our PCA-based cluster analysis differed significantly across key measures. The results revealed that the PCA-based clustering successfully identified important differences between the Upper Intermediate and Advanced groups. Specifically, the Advanced group scored significantly higher on vocabulary ($t(37)=-5.54$, $p<.001$) and IELTS reading ($t(37)=-6.20$, $p<.001$) than the Upper Intermediate group. These findings highlight that vocabulary size and reading proficiency are crucial indicators of L2 proficiency, effectively distinguishing between the two clusters.

The PCA-based clustering method contrasts with what would have been the outcome if we had relied solely on the traditional IELTS cut-off method. While the IELTS-based grouping would have lumped participants into broad categories that did not fully capture their actual proficiency, the PCA-based approach provides a more refined classification. By considering a broader range of variables and patterns, the PCA clustering method uncovers meaningful distinctions in participants' language skills that are more aligned with their cognitive processing, as shown by our eye-tracking data.

Overall, the significant differences identified in vocabulary and reading scores underline the superiority of the PCA-based method over the traditional IELTS cut-off. The PCA approach offers a more accurate and nuanced representation of participants' proficiency, reflecting the complexity of bilingual language development far better than the rigid cut-off scores used in IELTS-based groupings.

3.2 Statistical modelling of first fixation durations

To analyse our dataset of 80,191 data points focusing on First Fixation Duration (FirstFixDur) in reading, we utilised Generalised Additive Mixed Models (GAMMs). This approach allowed us to explore the non-linear relationships between key covariates such as word frequency and word length, and the primary dependent variable. GAMMs were particularly effective in capturing the complex-

ities of bilingual language processing, accommodating the dynamic nature of factors like L2 acquisition and exposure.

A significant enhancement in our analysis was the inclusion of a group variable, informed by PCA and NCA. This allowed us to accurately assess how word-level and text-level indicators affect different groups of readers. The final model retained only the independent variables that exhibited statistical significance, excluding the frequency of the preceding and following words, which did not significantly impact FFDs. The model analyses FFD revealing how word length and frequency affect FFDs for readers of different proficiency and exposure levels (group). Overall, the model explains 15.1% of the deviance in FirstFixDur, with an adjusted R-squared value of 0.14. The results of the model are summarised in Table 3.

Table 3. Generalised additive mixed model results for FFD

Predictor	EDF	Ref. DF	<i>F</i>	<i>p</i>
s(Wordlength_Prev): groupAdvanced	3.232	3.941	1.617	.152
s(Wordlength_Prev): groupNative	3.298	3.988	4.886	<.001
s(Wordlength_Prev): groupUpper_Intermediate	3.671	4.452	2.158	.073
s(Wordlength_N): groupAdvanced	5.989	6.922	13.100	<.001
s(Wordlength_N): groupNative	1.775	2.088	20.220	<.001
s(Wordlength_N): groupUpper_Intermediate	4.187	5.035	12.103	<.001
s(Wordlength_Next): groupAdvanced	1.895	2.347	5.846	.0017
s(Wordlength_Next): groupNative	1.829	2.244	1.730	.161
s(Wordlength_Next): groupUpper_Intermediate	2.309	2.859	6.406	<.001
s(Fre_log_N): groupAdvanced	4.331	5.189	19.161	<.001
s(Fre_log_N): groupNative	3.059	3.616	11.657	<.001
s(Fre_log_N): groupUpper_Intermediate	3.586	4.295	13.178	<.001
s(Scaled_Word_Story_Order)	1.040	1.059	3.860	.050
s(WordStory)	807.419	1370.000	1.511	<.001
s(Scaled_Word_Story_Order, Participant_ID, bs="fs")	195.648	710.000	12.411	<.001

Note. EDF=estimated degrees of freedom; Ref. DF=reference degrees of freedom; *F*=*F*-statistic; *p* indicates significance level.

The model includes group-specific smooth terms for WordLength.N-1 (Wordlength_Prev), WordLength.N (Wordlength_N), WordLength.N+1 (Wordlength_Next), and LogWordFreq.N using the by=group specification, allowing us to assess how the effects of these predictors differ across groups.

As summarised in Table 3, the length of the current word (WordLength.N) has a significant impact on fixation duration for all groups, with the largest effects observed for advanced learners. Moreover, L2 learners show distinct patterns when processing adjacent words compared to L1 readers, with differences in how word length affects the FFDs for the preceding and following words.

After initial exploration and testing various predictors, the best fitting model showed significant impacts from the length of the previous, current, and next words (WordLength.N-1, WordLength.N, WordLength.N+1) and the frequency of the currently fixated word (LogWordFreq.N), with no further contribution from the frequency of surrounding words (i.e., LogWordFreq.N-1, LogWordFreq.N+1). Additionally, the model accounts for how a word's position in the text (Scaled_Word_Story_Order) influences FFDs. Finally, the model includes a set of random factors: the effect of unique word-story combination (Word-Story), and individual readers' fluctuations in FFDs across the words and stories (Scaled_Word_Story_Order). These by-Participant factor smooths account for how the subject's performance changes over the experiment as a consequence of changes in attention, fatigue, or learning.

The model highlights distinct patterns regarding how word length and frequency impact FirstFixDur across different reader groups. All groups – L1 readers, and upper intermediate and advanced L2 learners – demonstrate a significant effect of the length of the currently fixated word on FirstFixDur (L1: $F=20.220$, $p<.001$; upper intermediate L2: $F=12.103$, $p<.001$; advanced L2: $F=13.100$, $p<.001$). However, L2 readers differ from L1 readers in how they process surrounding words: L1 readers are influenced significantly by the length of the previous word ($F=4.886$, $p<.001$) but not the next word ($F=1.730$, $p>.1$), whereas L2 learners, both upper intermediate and advanced, are affected by the length of the next word (upper intermediate L2: $F=6.406$, $p<.001$; advanced L2: $F=5.846$, $p<.001$) but not the previous word (upper intermediate L2: $F=2.158$, $p=.07$; advanced L2: $F=1.617$, $p>.1$). These findings are visually represented in Figures 4–7.

Figure 4 demonstrates that previous word length does not significantly affect FirstFixDur for the two L2 groups. For L1 readers, there is a negligible effect for very short previous words, but FFDs increases for words of four or more characters, indicating a potential spill-over effect on processing (Findelsberger et al., 2019).

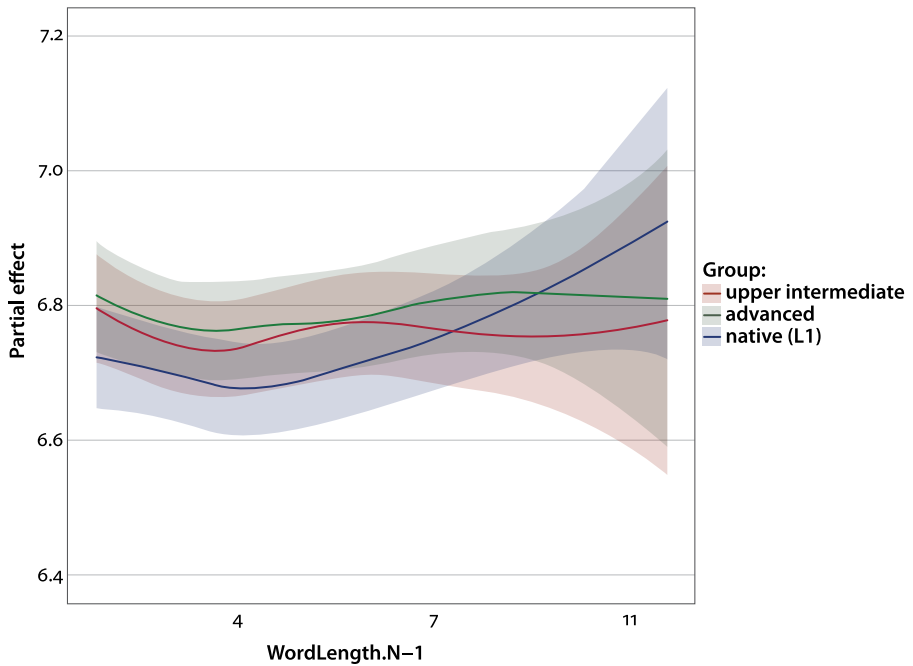


Figure 4. The effect of the length of the previously fixated word N-1 on the FirstFixDur on word N for two Chinese groups compared to the L1 group

Figure 5 illustrates that longer currently fixated words lead to increased First-FixDur, a trend consistent with previous reading studies indicating higher cognitive demands for processing longer words (Barton et al., 2014). However, the relationship appears nearly linear for L1 readers (blue line) but varies for the upper intermediate L2 group (red line), which experiences a plateau in fixation time across word lengths from 4 to 11 characters. This suggests upper intermediate readers might reach a threshold beyond which they are less sensitive to word length. In contrast, the advanced L2 group (green line) exhibits a nonlinear, more complex pattern, hinting at a shift towards native-like processing behaviour – a twisting curve (green line) around the linear trend (blue line).

Conversely, Figure 6 shows differing responses to the length of the next word among the groups: no significant effect for L1 readers (blue line), while both L2 groups exhibit shorter FFDs with varying degrees of deceleration – more pronounced for upper intermediate L2 readers (red line) and relatively stable for advanced L2 readers (green line). This might reflect anticipatory processing strategies, suggesting that less proficient L2 readers engage less in forward planning than their more advanced counterparts.

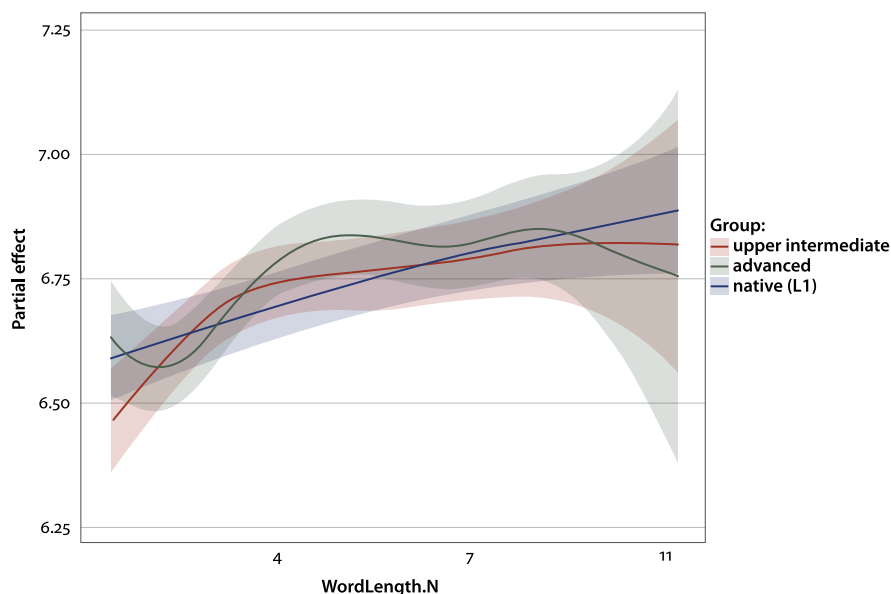


Figure 5. The effect of the length of the currently fixated word N on the FirstFixDur on word N for two Chinese groups compared to the L1 group

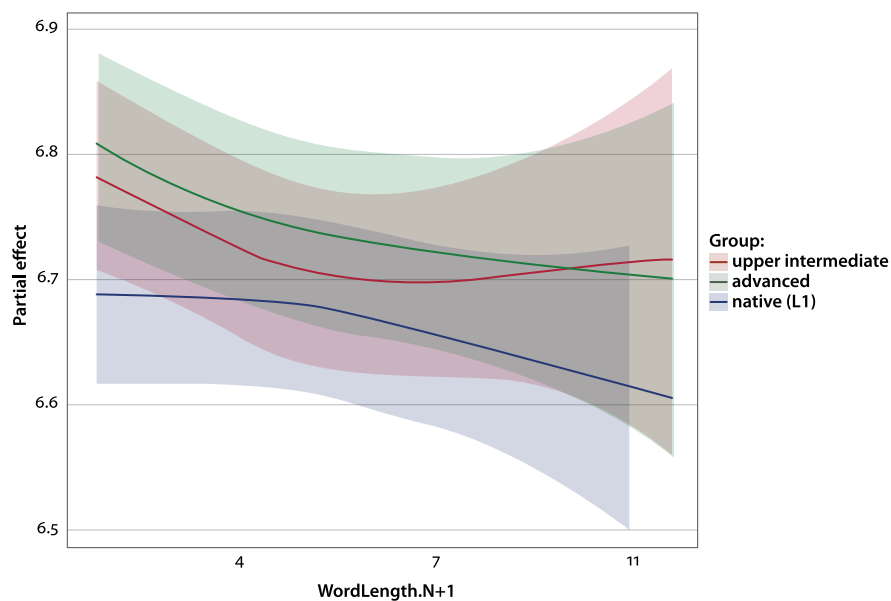


Figure 6. The effect of the length of the next fixated word N+1 on the FirstFixDur on word N for two Chinese groups compared to the L1 group

Figure 7 shows that all three groups of readers — L1, upper intermediate L2, and advanced L2 — exhibit a similar response to the frequency of the fixated word, contrary to the patterns observed with word length. There is a minimal to no effect for low-frequency words; however, once the frequency exceeds a certain threshold (around $\log(6)$, equivalent to frequency > 400), FirstFixDur begins to decrease as word frequency increases. This trend aligns with findings from numerous previous studies, both in eye-tracking and other experimental settings (Baayen et al., 2016; Divjak & Milin, 2020).

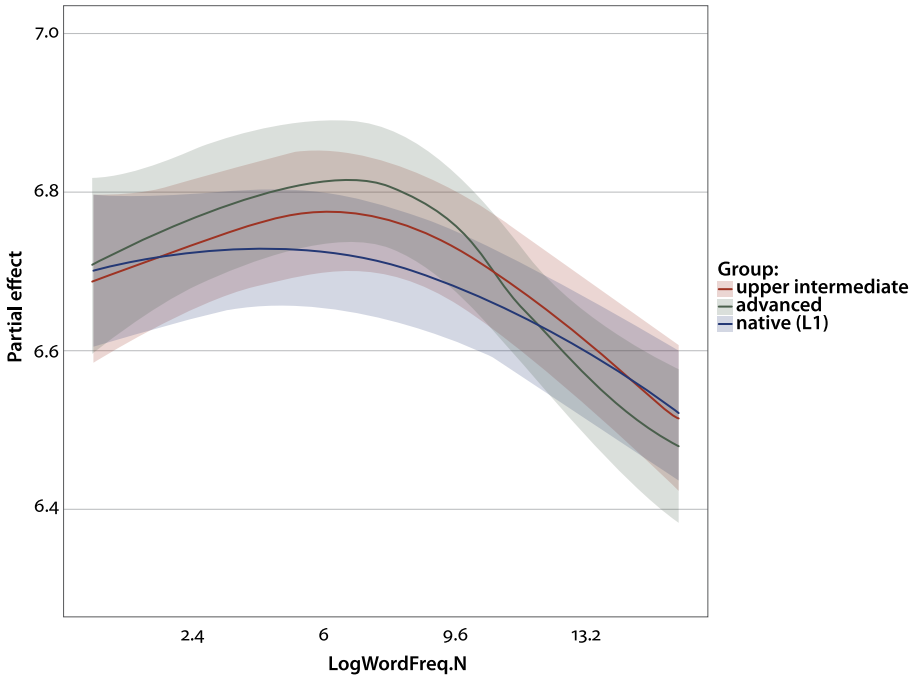


Figure 7. The effect of the frequency of the currently fixated word N on the FirstFixDur on word N for two Chinese groups compared to the L1 group

Significant effects of fixated word frequency are noted across all groups (L1: $F=11.657$, $p<.001$; upper intermediate L2: $F=13.178$, $p<.001$; advanced L2: $F=19.161$, $p<.001$), with a notably steeper decline for L2 than L1 readers. This suggests that proficient L2 readers are familiar with frequent English words and effectively utilise this knowledge during reading. The trend for advanced L2 readers (green line) shows a more rapid decrease in FFDs than that of upper intermediate L2 readers (red line), which may indicate a stronger familiarity with common English words.

4. Discussion

In this study we proposed a novel method for grouping L2 learners and explored how a novel, data-driven, and composite approach to grouping participants could yield deeper insights into the reading differences among L2 readers of varying proficiency and exposure levels.

We sought to create a method for effectively grouping participants in bilingual studies by combining self-reported data, vocabulary assessments, and information on second language exposure. The findings suggest that this multifaceted approach enhances our comprehension of bilingual language proficiency and reading capabilities. Employing PCA and NCA, we differentiated distinct groups within the bilingual participants, revealing substantial variability in proficiency and exposure, thus advancing beyond traditional grouping methods.

We then examined the impact of our innovative grouping method on the analysis of eye movement patterns in bilingual reading. Utilising GAMMs, we obtained detailed insights into the underlying cognitive processes of bilingualism, marked by diverse reading behaviours across different groups. This segment of the study emphasised the influence of elements such as word length and frequency on reading efficiency, thereby validating the effectiveness of our grouping strategy in enhancing the understanding of bilingual reading patterns.

Overall, our study highlights the nuanced interplay between word frequency, word length, reading proficiency and exposure, with distinct patterns emerging among learners of English at different proficiency levels. Notably, we observed a consistent nonlinear pattern of word frequency effect across all participant groups. Low-frequency words prompted longer FFDs, while high-frequency words resulted in shorter FFDs. This finding is in line with previous research conducted by Kliegl and colleagues (2006), illustrating the universal impact of word frequency on reading behaviours across languages and proficiency levels.

Moreover, our data revealed a complex nonlinear relationship between the length of the preceding word ($N-1$) and FFDs on the currently fixated word (N), closely paralleling the findings reported by Kliegl et al. (2006). Specifically, among L1 readers of English, when the preceding word was shorter than four letters, an increase in word length led to shorter FFDs on the subsequent word. However, for preceding words longer than four letters, an increase in word length corresponded with longer FFDs on the current word. This relationship was distinctly nonlinear for L2 readers, diverging from the linear trend observed for L1 English readers in our investigation.

Crucially, our analysis also uncovered that the advanced L2 group demonstrated a greater reliance on frequent English words, resonating with the observations made by Ashby et al. (2005). Their study investigated the eye movements

of highly skilled and average readers during silent reading, unveiling how reading skill modulates the effects of word frequency and predictability on the word recognition process. Similarly, our findings suggest that enhanced proficiency in a second language amplifies the capacity to exploit frequency information, potentially indicating more sophisticated lexical access and processing strategies among advanced L2 learners. This underscores the significance of language proficiency in refining reading efficiency and strategic approach, particularly in the context of English.

Moreover, we found that L1 readers are affected by the length of the previous word ($N-1$), while both groups of L2 readers are affected by a next word's length ($N+1$). As longer words typically require more effort for processing and integration, it is reasonable to speculate that the L1 readers use longer fixations to efficiently integrate information up to the currently fixated word; hence the longer fixations when $N-1$ word is longer too (Nilsson, 2012; Rayner & Liversedge, 2011). Conversely, L2 readers do not exhibit such a strategy but rather spend less time on the currently fixated word if the following word is longer. Thus, it seems that they invest in eye movement planning rather than in information processing and integration. Together, this shows a profoundly different approach in reading between L1 and L2 readers.

These findings would not have been obtained using a traditional IELTS-based participant grouping³, highlighting several key points for bilingualism research and cognitive psychology more generally. First, they affirm the necessity of a detailed approach to grouping participants, recognising that bilingual proficiency and exposure are complex and multifaceted. This method allows for deeper, more precise insights into bilingual language processing, moving beyond traditional and static categorisations. Second, the proposed approach together with the use of the eye-tracking experimental method not only advances our understanding of bilingualism but also shows the potential to guide the development of educational strategies tailored to the needs of bilingual individuals. Finally, the results enhance our understanding of how linguistic skills, proficiency levels, and language exposure collectively influence reading strategies, showcasing the adaptive nature of bilingual cognition and how bilinguals manage reading in different languages.

Our study paves the way for future research. There are opportunities to test the effectiveness of the grouping method in diverse linguistic and cultural settings to understand the universal and specific aspects of bilingual proficiency. Longitudinal studies could shed light on how bilingual reading strategies develop and change with varying levels of language exposure and proficiency. Moreover,

3. See Appendix 5 for the modelling results of the IELTS cut-off method. More discussion and comparison can be found in the online supplementary materials.

applying more sophisticated statistical analyses could unravel the complex interactions between cognitive mechanisms and linguistic factors in bilingual reading. The proposed method is straightforwardly transferable to other experimental techniques, suitable for specific research questions. Investigating additional variables, such as sociolinguistic features, individual learning styles, motivational factors, and specific language learning goals, may offer further insights into bilingual cognitive processes. Given the need for more refined participant grouping in bilingualism research, we aimed to evaluate the effectiveness of our PCA and NCA-based method against the traditional IELTS cut-off grouping. The PCA method, by integrating multiple proficiency indicators such as vocabulary test scores and time spent in the UK, offers a more comprehensive approach to capturing the multifaceted nature of bilingualism. In contrast, the IELTS cut-off approach, which is based solely on one-dimensional language test results, risks oversimplifying proficiency and missing subtle but significant differences in language exposure and usage.

The comparison between the two methods demonstrates that the PCA-based approach not only produces a more balanced distribution of participants but also reveals reading behaviour patterns that are not detected by the IELTS-based categorisation. The uneven distribution of participants in the IELTS-based method highlights its limitations in accurately reflecting the complexities of bilingual proficiency. The PCA method's inclusion of diverse variables allows for a more precise and meaningful segmentation, thereby improving the interpretation of cognitive processing data from eye-tracking. A detailed comparison of the modelling results from both methods is provided in the supplementary materials for further clarity and reference.

While this study provides valuable insights into bilingual reading strategies and proficiency grouping, several limitations should be noted. First, the generalisability of the findings is constrained by the homogeneity of the participant pool, consisting solely of Chinese–English bilinguals. Expanding the study to include participants from diverse linguistic and cultural backgrounds could reveal broader patterns in bilingual processing. Additionally, the study's reliance on eye-tracking data as the primary measure of reading behaviour, while effective, may not capture other cognitive processes involved in bilingual reading, such as memory or attention allocation. Future research could employ complementary methodologies, such as neuroimaging or verbal protocols, to provide a more comprehensive understanding of the cognitive mechanisms at play. Finally, although the use of PCA and NCA allowed for a more nuanced grouping of participants, these methods have inherent constraints, particularly in the subjective selection of the number of clusters. Alternative clustering techniques, such as hierarchical

models, could further refine participant classification and ensure more robust conclusions.

5. Conclusions

In conclusion, this study sheds light on the intricate nature of bilingual proficiency, employing an innovative approach to participant classification alongside an in-depth analysis of eye movement patterns. By emphasising the significance of acknowledging the varied experiences and competencies of bilingual individuals, our research provides key insights into the cognitive mechanisms that underpin bilingual language processing. As bilingualism research progresses, the methods and discoveries of this study promise to influence future research trajectories, deepen our grasp of the bilingual experience, and guide educational strategies and policies designed to support bilingual learners effectively.

Funding

Open Access publication of this article was funded through a Transformative Agreement with University of Birmingham.

Acknowledgements

We sincerely thank all participants who generously contributed their time to our study during the COVID-19 national lockdown in the UK in late 2020. Your support during this challenging period was invaluable to our research. We sincerely appreciate the insightful feedback from the anonymous reviewers and editors, which has greatly contributed to strengthening our manuscript. Special and deepest gratitude to James Marsden for his unwavering support throughout the writing and revision of this paper.

Declaration of competing interest

We have no known conflict of interest to disclose.

Data availability statement

Experimental data and code prepared for the study and statistical analyses performed for the purposes of the present paper are publicly accessible at the following location: https://osf.io/saemg/?view_only=of269a6c4b734fd2aob2doee919f3004. Additionally, the appendices referenced in the paper are available at this link.

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Publication history

Date received: 14 March 2024
Date accepted: 31 March 2025
Published online: 28 April 2025